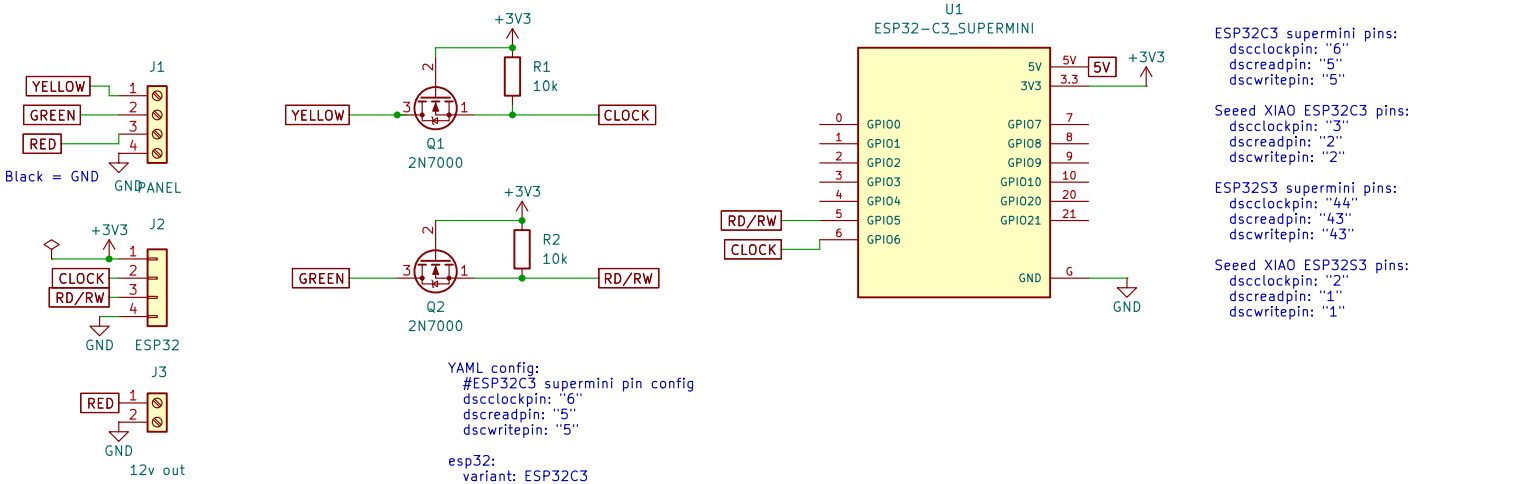
DSC interface – Mosfet version

J1: 3-position 5.08 mm screw terminal, 22-16 AWG alarm wire



ESP32C3 supermini pins:
  dscclockpin: "6"
  dscreadpin: "5"
  dscwritepin: "5"

Seeed XIAO ESP32C3 pins:
  dscclockpin: "3"
  dscreadpin: "2"
  dscwritepin: "2"

ESP32S3 supermini pins:
  dscclockpin: "44"
  dscreadpin: "43"
  dscwritepin: "43"

Seeed XIAO ESP32S3 pins:
  dscclockpin: "2"
  dscreadpin: "1"
  dscwritepin: "1"

YAML config:
  #ESP32C3 supermini pin config
  dscclockpin: "6"
  dscreadpin: "5"
  dscwritepin: "5"

esp32:
  variant: ESP32C3
  board: esp32-c3-devkitm-1

Notes:
i: resistors r1,r2 connection can be omitted as the internal gpio will be set as input_pullup
ii: q1,q2 can be substituted for other low power N-channel Mosfets. Be careful of pin locations as some are reversed such as with the bs170.
iii: the yellow and green lines from the panel already have builtin 10k pullup resistors so none are needed on the interface.
iv: j3 provides 12v output from the panel that can be used by an external step down to USB converter for powering the ESP.
  Powering the esp directly on the 5v pin is not recommended as it will conflict with the usb input if a cable is plugged in.